bFaaaP — Questions for H. Narusawa (#4)

Mechanical-layout reflection + motor successor + firmware-skeleton check

To: H. Narusawa   2026-06-26

Thank you for your replies. We reflected them and drafted a Pico firmware skeleton for the chosen 'approach A' (closed-loop NEMA17, STEP/DIR).

The firmware itself is handed over as a SEPARATE FILE:
  bFaaaP_autopro_pico_stepperA_draft_20260626.ino (an AI draft, untested — verify on the bench)

This PDF covers (1) the figure-reflection check and (2) the firmware confirmation points (TODOs to fill on the bench) + motor + CAD questions.

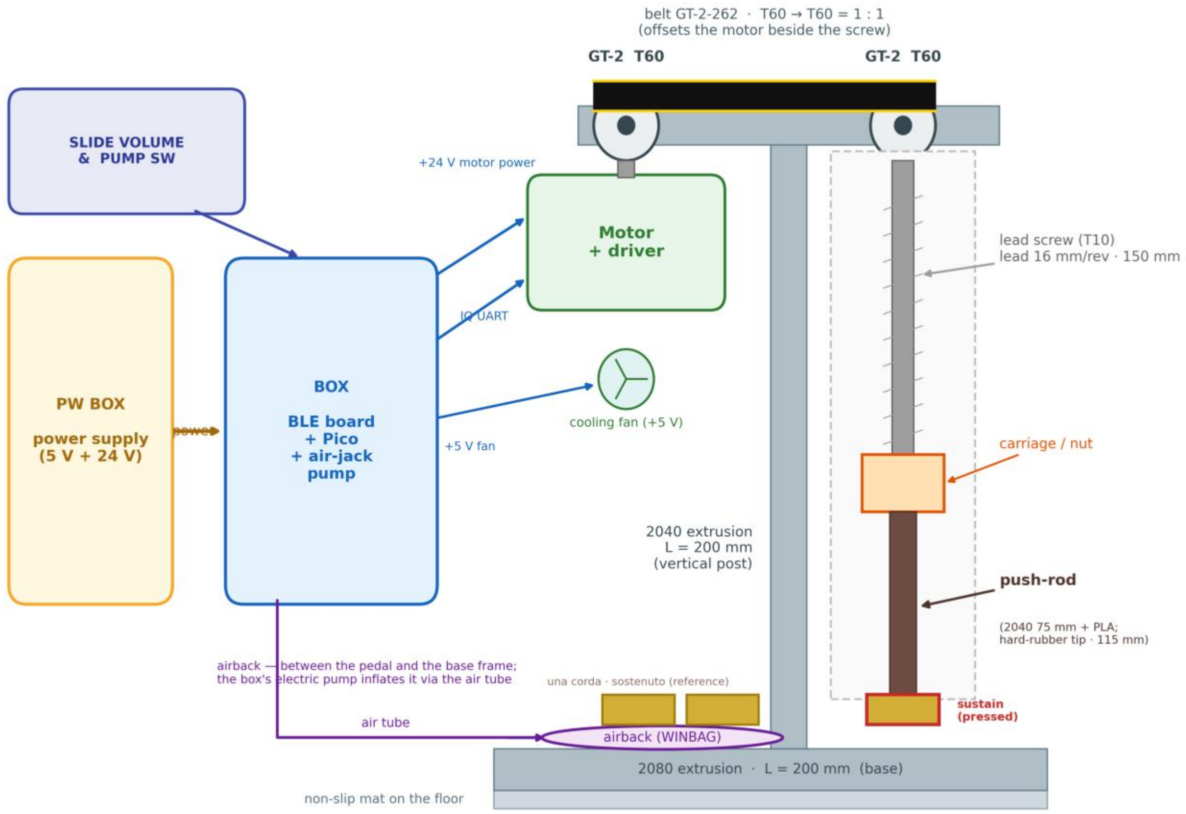
Already reflected (from your earlier replies):
- Mechanical layout: the 4 points (belt / airback position / 2080 base / non-slip mat)
- Motor successor: force-feedback 'approach A' (closed-loop)
- Firmware: a fresh skeleton that fixes the v052B STEP/DIR pin clash (GP13)

- This PDF reads fine on Windows (each page is rasterised; no fonts needed).

Contents: 1. mechanical-layout reflection   2. firmware / motor / CAD confirmations

1. Reflected your 4 points into the mechanical layout (please confirm)

bFaaaP Pro — mechanical layout (after the working unit's build)



Drawn by the bFaaaP project (CC BY 4.0) after the device co-author's (H. Narusawa) build + corrected wiring of the working unit. Schematic — proportions not to scale.

* Reflected 4 points

- (1) belt drawn as a real toothed belt (<- this point (1) is our interpretation)
- (2) airback placed BETWEEN the pedal and the base frame
- (3) '2080 extrusion L=200mm (base)' moved to the old 'una corda/sostenuto reference' spot
- (4) 'non-slip mat on the floor' placed at the base, matched to the base length
- (bottom stack: mat -> 2080 base -> airback -> pedals)

* Please confirm

- Is the belt interpretation right? If not, a corrected image lets us match it exactly.

2. Firmware / motor / CAD confirmations

A. Firmware (separate file) - fill in / confirm on the bench
  (1) driver load-read command (readLoad TODO): MKS SERVO42C/D serial/CAN, or the integrated driver's protocol. Which, and the comms spec? (a default ADC fallback runs on the bench.)
  (2) STEPS_PER_MM: microstep (e.g. 400) x 16 mm lead -> steps per 1 mm. Real value?
  (3) force thresholds (LOAD_THRESH_UP / DOWN): piano-dependent; bench values?
  (4) pins: STEP=GP14 / DIR=GP15 (fixes the v052B GP13 clash) / Serial2=GP4,GP5 (driver) / slider=A0,GP26 / pump=GP12 / buzzer moved to GP16. OK?
  (5) pressure sensing: HX711 (GP2/GP3) or ADC? (successor drops HX711, fixed 40 s inflate)
  (6) push direction sign: steps>0 = down (toward the pedal)?

B. Motor (approach A)
  - 'MKS SERVO42C/D + NEMA17' or 'integrated closed-loop NEMA17' - which? recommended torque/mode

C. CAD (you mentioned additions / replacements)
  - what is being added / replaced, and roughly when? we will reflect the final set into assembly/.

D. Tanaka's & Taguchi's reviews will be added to the thread / figure / docs as they arrive.

-- Reference (GitHub) --
  firmware: device-pro-acoustic/firmware/pico/bFaaaP_autopro_pico_stepperA_draft_20260626.ino
  structure: device-pro-acoustic/firmware/CODE-STRUCTURE.md
  figures/BOM: device-pro-acoustic/hardware/reference-design/ . device-pro-acoustic/assembly/